

From Region Proposals to Direct Regression: A Systematic Benchmark of Monocular Spacecraft Pose Estimation on SPEED and SPEED+

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Abstract—We report a systematic experimental study of 6-DoF monocular pose estimation for the Tango spacecraft on the SPEED and SPEED+ datasets. Five families are implemented and evaluated under a common protocol (translation error ET, rotation error ER, detection IoU, plus SPEED score where available): (i) SPN (Sharma & D’Amico) with AlexNet and ResNet-50 RPN variants, one-pass and two-pass full-resolution cropping; (ii) PVNet pixel-wise voting with PnP, across AlexNet/ResNet/ViT/Swin backbones, 8 vs. 11 keypoints, and SPEED to SPEED+ fine-tuning, with SAM ViT-H generated masks; (iii) HRNet-style deep landmark regression with nonlinear refinement (Bo Chen et al.); (iv) YOLOv11n detection at 640 px plus a YOLO to crop to ResNet-50 keypoint extension; (v) FastPose direct-regression Vision Transformer without PnP, plus a DETR-style detection front-end. The strongest detector is YOLOv11n (mean IoU 0.951 synthetic / 0.912 real). The strongest pose estimator is FastPose direct regression (mean ER $\approx 1.9^\circ$ synthetic / 2.4° real on the validation/real split used here). PVNet with 11 keypoints, SPEED pretraining and SPEED+ fine-tuning is the best PnP-based model (mean ER 7.5° real). SPN two-pass with a ResNet-50 RPN reaches mean IoU 0.84 synthetic / 0.77 real. Transformer backbones (ViT, Swin) underperform convolutional backbones for dense voting in this setting. This report documents architectures, data pipelines, all per-variant results including negative results, and domain-gap observations. It is a portfolio technical report, not a peer-reviewed publication; all baseline ideas are credited to their original authors.

Index Terms—spacecraft pose estimation, SPEED, SPEED+, 6-DoF pose, SPN, PVNet, YOLO, vision transformer

Code: notebooks in `spn, pvnet, hrnet, yolo, fastpose`; metrics in `src/common/metrics.py`. Datasets are not redistributed. See Appendix.

I. INTRODUCTION

Autonomous rendezvous requires estimating the 6-DoF pose (rotation $R \in SO(3)$, translation $t \in \mathbb{R}^3$) of a non-cooperative target from a single monocular image. The SPEED [1] and SPEED+ [2] datasets with the Tango spacecraft are the standard benchmark, with an explicit synthetic-to-real domain gap (synthetic renders vs. lightbox/sunlamp testbed images).

This report consolidates work done in 2025–2026, in chronological order:

- 1) SPN re-implementation and two-pass improvement.
- 2) PVNet adaptation to SPEED, including mask synthesis, backbone and keypoint ablations, and domain transfer.
- 3) HRNet-style landmark regression (Bo Chen et al.).
- 4) YOLO detection and YOLO-keypoint extension.
- 5) FastPose direct regression (last and best model).

TABLE I

SPEED AND SPEED+ SPLITS (FROM THE FASTPOSE NOTES). S = SPEED, S+ = SPEED+, SYN = SYNTHETIC, LB = LIGHTBOX, SL = SUNLAMP.

Split	S syn	S real	S+ syn	S+ lb	S+ sl
Train	12000	5	47966	–	–
Val	–	–	11994	–	–
Test	2998	300	–	6740	2791

The contribution is *not* a new state-of-the-art method. It is a controlled comparison of representative paradigms (proposal + classification/regression, dense voting + PnP, heatmap regression, single-shot detection, direct Transformer regression) with unified metrics, plus practical findings on cropping, masks, keypoint count, and augmentation. Negative results are reported as-is.

II. DATASETS, SPLITS AND METRICS

a) *SPEED* [1].: $\sim 12\,000$ synthetic 1920×1200 8-bit monochrome images with quaternion (scalar-first) + translation labels and known intrinsics, plus a small labelled real set and unlabelled test splits. The split used here is 9 600 train / 2 400 validation from the 12 000 synthetic images, plus evaluation on real images.

b) *SPEED+* [2].: Extends SPEED with synthetic, *lightbox* and *sunlamp* domain-gap splits ($\sim 47\,966$ synthetic train, 11 994 synthetic val, 6 740 lightbox, 2 791 sunlamp in the author’s notes). PVNet fine-tuning experiments use SPEED+; other experiments report synthetic vs. real.

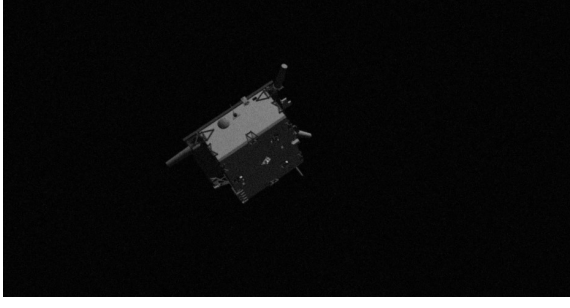
c) *Keypoints and camera*.: 11 Tango 3D points (`tangoPoints.mat`) and intrinsics/distortion (`camera.json`) are used throughout. Early PVNet ablations use 8 keypoints. 2D projections use the standard rigid transform + distortion + K pipeline (cf. T. Park’s SPEED code).

d) *Metrics*.: For predicted (q_{pred}, t_{pred}) vs. ground truth (q_{gt}, t_{gt}) :

$$ET = t_{pred} - t_{gt},$$

$$|ET| = \|ET\|_2 \text{ [m]}, \quad ER = \angle(R_{pred} R_{gt}^T) \text{ [deg]}, \quad (1)$$

where $R(\cdot)$ is the quaternion-to-DCM map and the angle is from $\text{trace}(R_{rel})$. Detection is measured by box IoU in $[x_1, y_1, x_2, y_2]$ format. We report mean and median. The official SPEED score $S = ER + |ET|/\|t\|$ (with thresholds) is reported only for the FastPose 310-epoch checkpoint where it



```
{
  "filename": "img000187real.jpg",
  "q_vbs2tango": [0.135, 0.837,
    -0.350, -0.398],
  "r_Vo2To_vbs_true":
    [0.206, 0.269, 3.297]
}
```

Fig. 1. Dataset example: top, raw Tango image; bottom, corresponding JSON pose label (quaternion + translation).

was computed; earlier runs report ET/ER/IoU only. This is a limitation noted in Sec. VI.

III. METHODS

A. SPN: proposal + attitude classification and position regression

Following Sharma & D’Amico [3]: an AlexNet-style backbone with Region Proposal Network (RPN) + Fast R-CNN head predicts a bounding box; parallel FC branches classify discrete attitude bins (uniform random quaternions, nearest-neighbour weighting, cf. the original MATLAB code) and regress position. Variants explored:

- **SPN-RPN (AlexNet):** baseline detector on projected-keypoint boxes.
- **SPNV100:** one-pass SPN (crop 224×224 directly with predicted box).
- **TPark baseline:** SPN trained on ground-truth crops (IoU = 1 by construction), isolating pose-head performance.
- **SPNV200:** SPNV100 initialised from TPark convolutional weights + hyperparameter changes.
- **SPNV300:** joint training with $loss = loss_{class} + 10 \cdot loss_{regress} + rpn_{loss} + roi_{loss}$.
- **SPNV400 (two-pass):** detect on downsampled image, then crop the *original full-resolution* 1920×1200 image and run SPN again. This avoids the information loss of cropping the 224×224 feature map and is the key SPN improvement here.
- **SPN-Advanced:** ResNet-50 RPN backbone replacing AlexNet, paired with the SPNV400 pose head.

B. PVNet: dense voting + PnP

Following Peng et al. [4]: a fully convolutional network predicts per-pixel unit-vector fields pointing to each 3D keypoint plus a segmentation mask; keypoints are voted (RANSAC-style) and pose is solved with `cv2.solvePnP` (iterative / Levenberg-Marquardt).

SPEED provides no masks, so masks are synthesised with Segment Anything (SAM ViT-H [10]) prompted by projected keypoints (PVNet segmentor notebook); ViT-B ablations and SPEED+ mask splits were also generated. Experiments:

- Backbones: AlexNet (PVNet8SPN), ResNet (SPEED8/11), ViT (ViTPose), Swin Transformer (Swin-PVNet), Swin segmentation decoder (Swin-Seg).
- Keypoints: 8 vs. 11.
- Domain: SPEED-only, SPEED+-only, SPEED pretrain + SPEED+ fine-tune.

Losses are smooth-L1 on voting vectors with mask weighting, plus segmentation loss.

C. HRNet: landmark regression + nonlinear refinement

Following Bo Chen et al. [5] with HRNet [6] backbone: 768×768 input, 11 joints, Gaussian heatmaps ($\sigma = 2$), top-k / OHKM options, followed by nonlinear pose refinement. Config: stage channels [32,64,128,256], rotation factor 30, scale factor 0.25 (see the HRNet training logs). Training completed but full test evaluation was not consolidated; results are reported as training logs only.

D. YOLO detection and YOLO-keypoint extension

Ultralytics YOLO [7] (YOLOv11n / YOLOv12n init) trained at 640px for 120 epochs (batch 32, $2 \times$ GPU, AMP, mosaic close 10, seed 0) on a YOLO-format SPEED conversion (see the YOLO dataset notebook). A second model (YOLO-extend) crops with the predicted YOLO box and regresses keypoints through a ResNet-50 + convolutional head.

E. FastPose: direct Transformer regression + DETR front-end

Following Ancy et al. FastPose-ViT [8] (code and geometric parameterisation provided by the authors, adapted to SPEED here). No PnP at inference.

a) *Crop.*: A squared crop (15% padding, zero-pad at borders, resize to 384×384) is derived from projected model keypoints. A separate lightweight DETR-style detector (LW-DETR: ResNet-18 + Transformer encoder-decoder with a single object query, MLP box head, $\lambda_{L1} = 5$, $\lambda_{GIoU} = 2$, 70 epochs) is trained to provide boxes at test time (mean IoU 0.88 synthetic / 0.86 real). Ground-truth boxes are generated by projecting the 3D wireframe (cf. T. Park’s code):

$$\begin{aligned} X_{cam} &= R(q) X_w + t, \\ B_{gt} &= [\min u, \min v, \max u, \max v] \pm \text{margin}, \end{aligned} \quad (2)$$

with perspective projection, radial/tangential distortion $\mathcal{D}(\cdot)$, and mapping $u = K[x_n, 1]^T$; margin is 5% for detection and 15% (squared) for FastPose crops.

b) *Parameterisation.*: The network predicts *apparent* orientation (the orientation a centred object would need to look the same) in continuous 6D form [9] plus a crop-normalised translation $U = (U_x, U_y, U_z)$ invariant to crop scale and intrinsics. Global R is recovered by a corrective rotation ΔR from the translation direction; see the implementation notes

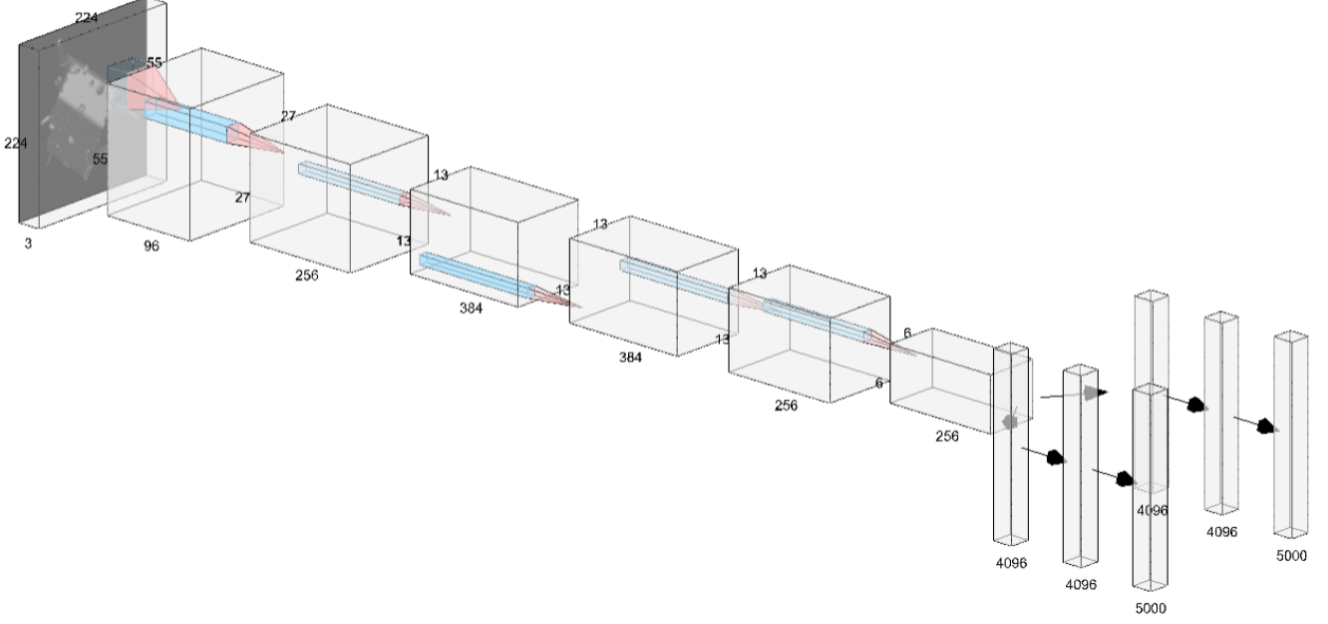


Fig. 2. SPN convolutional backbone used here (AlexNet-style, 224×224 Tango crop). Original diagram.

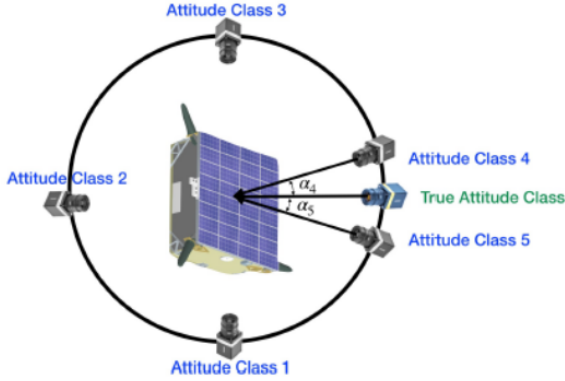


Fig. 3. Discrete attitude classes. The true attitude is expressed as a weighted combination of nearest classes. Adapted from Sharma & D’Amico [3].

for full equations. In short, the 6D output is orthonormalised by Gram-Schmidt,

$$\begin{aligned} r_1 &= \hat{r}_1 / \|\hat{r}_1\|, \\ r_2 &= (\hat{r}_2 - (r_1 \cdot \hat{r}_2)r_1) / \|\hat{r}_2 - (r_1 \cdot \hat{r}_2)r_1\|, \\ r_3 &= r_1 \times r_2, \quad R = \Delta R^T R'_{pred}, \end{aligned} \quad (3)$$

and translation is regressed in normalised crop coordinates,

$$\begin{aligned} U_x &= (f_x X / Z + c_x - b_x) / w, \\ U_y &= (f_y Y / Z + c_y - b_y) / h, \\ U_z &= (1/s_x + 1/s_y) Z / 2, \end{aligned} \quad (4)$$

with (b_x, b_y, w, h) the crop centre/size and s_x, s_y full-image to crop scale ratios.

c) Architecture and optimisation.: ViT-B/16 pretrained (SWAG), 384×384 input, 768-d feature to 9-d output (t_{rel}, r_{6d}) . MSE on rotation + translation. Hybrid optimiser: Muon (Newton-Schulz orthogonalisation, 5 iterations) for hidden 2D weights, AdamW for embeddings/biases/norms/head; AMP; cosine annealing with warm restarts ($T_0 = 10$, $T_{mult} = 2$). Augmentation: in-plane rotation ($\pm 20^\circ$ or 160–200°, $p = 0.5$, labels updated), box jitter ($\pm 10\%$), Albumentations photometric distortions (colour jitter, blur/noise, JPEG 85–100). Note: pixel augmentations help real transfer but hurt synthetic-only scores (ablated in [8]).

IV. IMPLEMENTATION DETAILS

Runs used Kaggle 2×GPU. The public repo remaps configs to the data folder (see Appendix). Shared metric helpers are in the common metrics module. YOLO curves and weights are kept under the YOLO training metrics folder; FastPose checkpoints (~660MB) are Kaggle kernel outputs, not in git. FastPose notebooks are committed with cell outputs so tables and plots render on GitHub.

V. RESULTS

All numbers below are taken from the per-folder test notebooks/READMEs (mean/median on synthetic and real splits as defined there). They are comparable within this report but not directly comparable to official Kelvin leaderboard scores, since splits and the SPEED score differ (Sec. II).

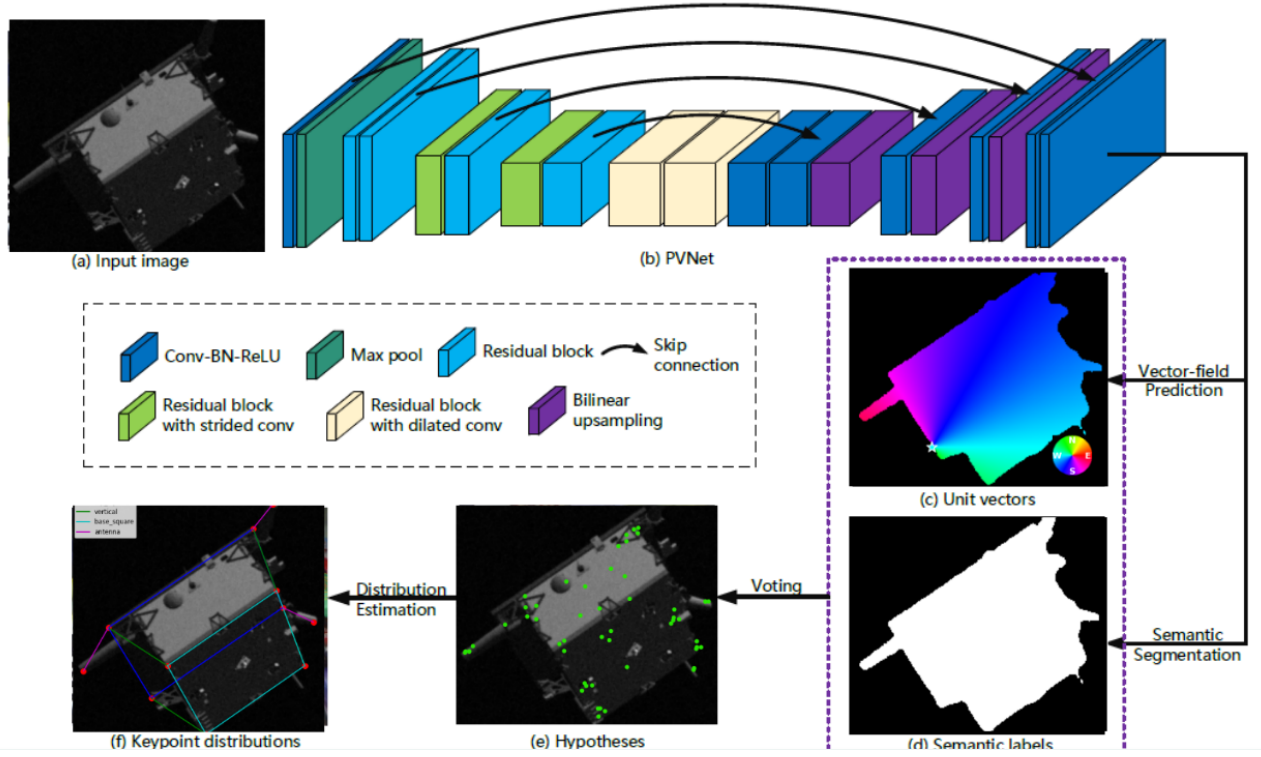


Fig. 4. PVNet pixel-wise voting pipeline with Tango images (input, voting backbone, unit-vector field, mask, hypotheses, keypoint distributions). Adapted from Peng et al. [4]; satellite panels are the author’s replacement for LineMOD.

TABLE II
SPN VARIANTS. IOU, TRANSLATION MAGNITUDE $|ET|$ (M), ROTATION ERROR ER (DEG). TPARK USES GT CROPS SO IOU = 1 BY CONSTRUCTION.

Variant	Synthetic			Real		
	IoU $\uparrow$	$ ET \downarrow$	$ER \downarrow$	IoU $\uparrow$	$ ET \downarrow$	$ER \downarrow$
V100 (AlexNet, 1-pass)	0.78	61.99	57.5	0.70	17.44	146.0
TPark (GT crop)	1.00	1.00	5.86	1.00	0.59	58.1
V200 (TPark init)	0.77	61.97	61.3	0.70	17.69	153.4
V300 (joint RPN)	0.72	61.60	45.3	0.70	18.26	128.9
V400 (2-pass full-res)	0.77	0.96	20.6	0.70	0.58	88.7
Adv (R50 RPN + 2-pass)	0.84	0.87	14.7	0.77	0.60	78.8

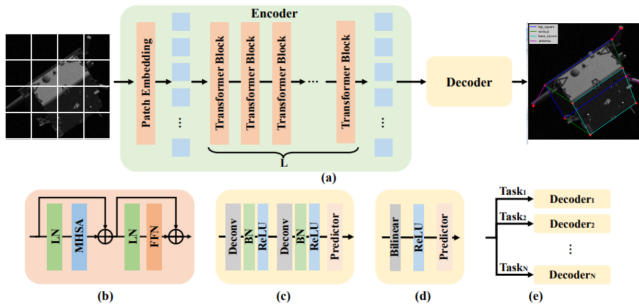


Fig. 5. ViT encoder-decoder pose head used for the ViT experiments, shown with a Tango crop. (a) patch embedding + Transformer encoder + decoder; (b)–(e) block and decoder variants. Author’s adaptation with satellite images.

A. SPN

Takeaway: one-pass SPN fails on translation (cropping the 224×224 map destroys scale). Two-pass full-resolution cropping cuts $|ET|$ from ~ 62 m to < 1 m synthetic. ResNet-50 RPN further improves IoU and rotation. Real-domain rotation remains poor for all SPN variants (69 – 175° median), motivating voting and regression approaches.

B. PVNet

Takeaways: (1) 11 keypoints strongly beat 8 (median ER 12.0° vs. 16.1° synthetic; 12.2° vs. 148.0° real). (2) SPEED pretraining + SPEED+ fine-tuning gives the best real rotation (7.49° mean, 7.93° median). (3) ViT/Swin backbones underperform ResNet for dense voting here (consistent with ViT being weak at dense prediction without a dense decoder). Median translation remains large (~ 8 m synthetic) because far-range samples dominate the mean.

TABLE III
PVNET VARIANTS (MEAN $|ET|$ IN M, MEAN ER IN DEG). BEST PVNET IN BOLD.

Variant	$ ET $ syn.	ER syn.	$ ET $ real	ER real
AlexNet, 8 kpt	10.63	112.3	2.20	110.4
ResNet, 8 kpt	7.26	66.0	2.61	123.2
ResNet, 11 kpt	9.85	45.6	0.97	13.37
R11, SPEED+ only	10.03	54.6	2.20	46.6
R11, SPEED $\rightarrow$ SPEED+ ft	9.42	31.0	1.29	7.49
ViT backbone	10.38	117.1	4.34	148.2
Swin-PVNet	10.85	87.4	3.74	93.3
Swin-Seg	10.27	68.0	2.89	136.6

TABLE IV
DETECTION RESULTS (IoU).

Model	IoU synthetic	IoU real
YOLOv11n	0.951 (med. 0.965)	0.912 (med. 0.910)
DETR front-end	0.882 (med. 0.913)	0.858 (med. 0.822)

TABLE V
POSE RESULTS (S = SYNTHETIC, R = REAL).

Model	$ ET $ s/r (m)	ER s/r ($^{\circ}$)
YOLO $\rightarrow$ R50 kpt	6.01 / 4.85	43.8 / 130.8
FastPose-ViT	0.043 / 0.035	1.91 / 2.43

TABLE VI
FASTPOSE 310-EPOCH CHECKPOINT VS. REPORTED FASTPOSE-ViT RESULTS (MEAN ERRORS).

Metric	Paper	Ours syn	Ours real
Mean $ ET $ (m)	0.0221	0.0046	0.0280
Mean ER ($^{\circ}$)	0.4183	1.2238	4.7625
SPEED score	—	0.0257	0.0943

C. YOLO and FastPose

FastPose numbers are from the FastPose test notebook (9600/2400 val + real). The implementation notes report a 310-epoch checkpoint at 0.0046 m / 1.22 $^{\circ}$ synthetic and 0.028 m / 4.76 $^{\circ}$ real (SPEED scores 0.026/0.094).

Three likely reasons for the gap, from the notes: (1) near-perfect keypoint-derived crops lower translation vs. detector crops; (2) training on 9600 vs. 12000+5 images hurts rotation; (3) pixel augmentations (blur, noise, JPEG) help real transfer but hurt synthetic-only rotation.

D. Best-of-each summary

YOLO training converges cleanly (120 epochs: precision/recall $\approx$ 0.999, mAP50 0.995, mAP50-95 0.963). The YOLO-keypoint extension overfits synthetic rotation (median ER 9.6 $^{\circ}$ synthetic vs. 156.9 $^{\circ}$ real). FastPose direct regression is the only method with single-digit real rotation and centimetre-level translation, at the cost of an 86 M-parameter ViT-B backbone. Table VII summarises the best variant per family.

TABLE VII
BEST VARIANT PER FAMILY (MEAN ERRORS; SYNTHETIC / REAL).

Family	$ ET $ s/r (m)	ER s/r (deg)	PnP?	Notes
SPN-Adv 2-pass	0.87/0.60	14.7/78.8	no	IoU 0.84/0.77
PVNet-R11 + ft	9.42/1.29	31.0/7.5	yes	ResNet + voting
HRNet-BoChen	(logs only)	(logs only)	yes	HRNet-W32
YOLOv11n	—	—	—	IoU 0.95/0.91
FastPose-ViT	0.043/0.035	1.9/2.4	no	86 M ViT-B

VI. DISCUSSION

a) *Why FastPose wins here.*: It avoids the two failure modes of the other pipelines: (a) quantised attitude bins and scale loss in SPN, (b) noisy dense votes + PnP sensitivity in PVNet, especially under domain shift. The apparent-rotation + normalised-translation parameterisation makes the regression target crop-invariant, and the ViT capacity plus aggressive augmentation absorbs appearance shift. Inference is single-pass (no iterative PnP).

b) *Domain gap.*: All PnP-based methods degrade sharply synthetic $\rightarrow$ real in rotation (e.g. SPNV100 median ER 19.5 $^{\circ} \rightarrow$ 147.9 $^{\circ}$; PVNet-R8 16.1 $^{\circ} \rightarrow$ 148.0 $^{\circ}$). SPEED pretrain + SPEED+ fine-tune and photometric augmentation partially close the gap. Pixel augmentation hurts synthetic-only scores but helps real transfer, consistent with [8].

c) *Crop quality dominates translation.*: SPN $|ET|$ drops two orders of magnitude when cropping full resolution; FastPose translation is best when given near-perfect keypoint-derived crops, and degrades with detector boxes. Detection (YOLO 0.95/0.91 IoU) is thus not a solved side-task but a first-order pose error source.

d) *Backbones for voting.*: ResNet > AlexNet $\gg$ ViT/Swin for PVNet voting fields in these runs. Likely causes: insufficient dense decoder for Transformers, data scale, and hyperparameter reuse across backbones. Not a general claim about Transformers (FastPose-ViT itself is best overall for direct regression).

e) *Limitations (read before reusing numbers).*:

- Splits and SPEED scores are inconsistent across runs (val vs. test, synthetic/real vs. lightbox/sunlamp). Future work should report the official SPEED score on fixed SPEED+ lightbox/sunlamp.
- FastPose has two reported checkpoints (test notebook vs. 310-epoch notes PDF); both show the same ranking but absolute numbers differ. The checkpoint ($\sim$ 660 MB) is a Kaggle output, not in git.
- HRNet evaluation is incomplete (training logs only).
- No latency/FLOP table was measured; PnP vs. direct-regression cost is qualitative here.
- Far-range samples inflate mean $|ET|$; median and range-stratified errors should be reported alongside.

VII. CONCLUSION AND FUTURE WORK

Five paradigms were compared on the same spacecraft problem with shared metrics. Practically: use YOLO-class detectors for boxes, use full-resolution two-pass crops if a

PnP pipeline is required, prefer 11 keypoints and SPEED $\rightarrow$ SPEED+ fine-tuning for voting methods, and prefer direct apparent-pose regression (FastPose-style) for best accuracy without PnP. Open items: unified SPEED-score evaluation on lightbox/sunlamp, detector-in-the-loop FastPose scoring, HRNet test completion, and efficiency benchmarking for onboard deployment.

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APPENDIX

Per-folder READMEs hold per-notebook tables. Layout:

- `spn`: SPN variants and tests.
- `pvnet`: PVNet variants and segmentor.
- `hrnet`: landmark regression and training logs.
- `yolo`: dataset, training, extension and metrics.
- `fastpose`: training, testing, DETR pair and notes.
- `docs`, `data`, `src/common`: presentation, references, data layout and metrics.

Datasets (~10 GB) and SAM/YOLO/FastPose weights are not in git; Kaggle mirrors (see the Kaggle doc in the repo) give repro sources. Original runs used Kaggle 2 $\times$ GPU.